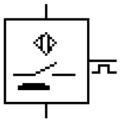
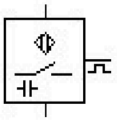
Sensors

Proximity Sensors

- Proximity sensors check for the presence of something a short distance away from the sensor.
- There are two types covered:
- Inductive Proximity Sensor
  - detects both ferrous and nonferrous metals
  - the actual sensing distance is determined by the rating sensing distance, callibrated by a 1mm thick square steel plate, multiplied by the correction factor and the manufacturer tolerance.

$$\text{Actual Sensing Distance} = \text{Rated Sensing Distance} \times \text{Correction Factor} \times \text{Manufacturer Tolerance}$$

  - the correction factor is dependent on the material being detected.
  - inductive proximity sensors and generally inexpensive, rugged, reliable and have a long service life
  - used in applications like:
    - detecting presence of metal bottle caps
    - detectign number of teeth passing through to measure rotational speed or position
- Capacitive Proximity Sensor
  - detects almost any object with electrostatic fields
  - sensing distance affected by sensor diameter and target size, as well as the dielectric constant (higher $\rightarrow$ better)
  - used in applications like:
    - non contact liquid level sensor
    - counting non metallic parts
  - However, if a film or dust or dirt forms on surface, the sensor may not work as well

Characteristic	Inductive Sensor 	Capacitive Sensor 
Suitable for detecting	Metals	Metals and non-metals
Detecting distance	1 to 30 mm (Long range: 60mm)	3 to 30 mm
Switching frequency	250 to 2 kHz	10 to 250 Hz
Size	Smaller	Larger
Sensitivity Adjustment	No	Yes

Photoelectric Sensors

- has two parts, emitter and reciever.
- The reciever can be setup in two modes, Dark-On, when the input drops below a threshold, and Light-On, when the input rises above a threshold.
- There are three common setups:
- Through Beam
  - emitter placed opposite reciever
  - beam is broken when opaque object is placed
  - for long range applications (~40m, e.g. garage door sensor)

alignment is an issue

Retro-Reflective

emitter embedded with receiver, reflector placed opposite

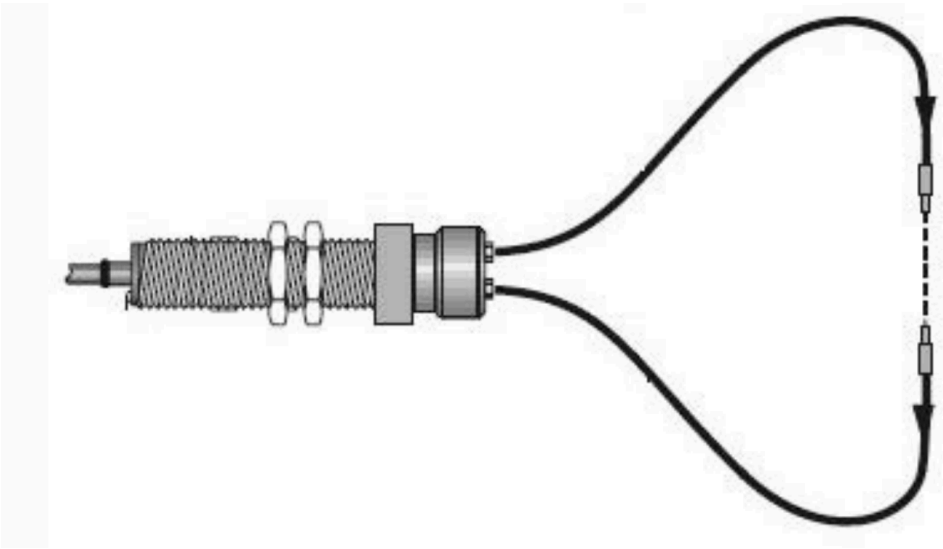
medium range applications (~4m, e.g large production lines)

Diffuse Reflective

emitter embedded with receiver, light reflects off detected object.

short range applications (~1m)

does not work well for dark or dull obejcts

susceptible to background light affecting sensor

another application for through beam is for fiber optic cable alignment, for small parts that need to be aligned or detected: